

unit - II

Cataloging and Indexing

⇒ History and objectives of Indexing in IRS:-

Indexing is the process of organizing & representing documents by extracting important words (called index terms or keywords) so, that users can quickly & accurately retrieve relevant information.

Indexing is the process of creating an organized list of keywords that helps users find required documents quickly without searching the entire collection.

History of Indexing:-

The concept of Indexing has evolved over many centuries.

Stage 1: Ancient period:-

- Information was stored in handwritten manuscripts
- Libraries maintained handwritten catalogues
- Documents were arranged alphabetically or by subject.
- Retrieval was slow because everything was done manually.

* Mathematics

* philosophy

* Medicine

Stage 2: printed Book era (15th century)
After the invention of the printing
Press the no. of books increased rapidly.

Problems:- huge no. of books
Difficult to locate information

Solution:- publishers started adding

- * Table of contents
- * Alphabetical index
- * Subject Index

These Indexes helped readers locate
information quickly.

Stage 3:- Library Indexing (19th-20th century)

As Libraries expanded professional
Libraries developed indexing standards

Important developments

- Card catalogues
- Subject Headings
- Classification sm

Evolution of Indexing:-

Period	Method
Ancient age	Handwritten catalogues
Printing era	Book indexes
Library era	Card catalogues & Subject Indexing
Computer era	Electronic Indexing
Modern era	Automatic Search engine Indexing

Objectives of Indexing

- 1) fast information Retrieval
- 2) Reduce Search Time
- 3) Improve Retrieval accuracy
- 4) organize Large collections
- 5) Support Keyword searching
- 6) Reduce storage Access
- 7) Improve Search performance
- 8) Increase precision
- 9) Increase Recall
- 10) Support multiple search techniques
- 11) Eliminate Duplicate searching
- 12) Efficient Resource utilization
- 13) Support automatic Retrieval
- 14) Improve user satisfaction
- 15) Handling bigdata

The main purpose of Indexing is to improve the efficiency of an Information Retrieval system.

- 1) fast Information Retrieval

The primary objective is to retrieve information quickly. without Indexing every document must be searched. with Indexing only the Index is searched.

Ex:- Instead of searching 10 lakhs documents the system searches only the Index.

- 2) Reduces search time:-

Indexing minimizes the time required to locate information.

Benefits:-

- * Immediate access
- * Faster response
- * Efficient searching

3) Improve Retrieval Accuracy:-
Indexing ensures that only relevant documents are retrieved.

Advantages:-

- * Less irrelevant information
- * Better search quality
- * Increased precision.

4) organize large collections:

- Large databases contain millions of documents
- Indexing organizes them systematically

Organization methods

→ Alphabetical

→ subject - wise

→ category - wise

→ Keyword - wise

5) Support keyword searching:-

users search using keywords

The index maps keywords to documents.

Ex:- keyword → "Blockchain"

Returns:-

Document 2

Document 15

Document 27

6. Reduce storage Access:-
Instead of opening every file. The
System checks only the index

Result:- * Reduced disk access
* faster processing

Improve search performance

Performance is measured by

* speed

* Accuracy

* Response time

Indexing improves all these parameters

8) Increase precision:-
Precision means retrieving only relevant
documents.

Formula:-
$$\text{Precision} = \frac{\text{Relevant Retrieved Documents}}{\text{Total retrieved documents}}$$

Higher precision means fewer unwanted
documents.

9) Increase Recall:-

Recall means retrieving all relevant
documents.

Formula:-
$$\text{Recall} = \frac{\text{Relevant retrieved documents}}{\text{Total relevant documents}}$$

* Good Indexing improves recall

10) Support multiple search Techniques:-

Indexing enables

→ keyword search

→ Boolean search

- Phrase search
- Wildcard search
- Proximity search

Indexing process:-

Introduction:- Information retrieval systems manage huge collection of documents such as books, journals, newspapers, research papers, reports, webpages and digital files.

⇒ The no. of documents increases, search for the required information becomes difficult and time-consuming.

To solve this problem indexing is used.

⇒ Indexing creates a structured list of important terms from documents so, that the required information can be found quickly without searching the entire document.

Indexing process is the process of identifying important concepts or keywords from a document and storing them in an index to enable fast & accurate retrieval of information.
(or)

Indexing is the process of representing the subject content of a document.

Why is Indexing process Important:-

When an organization has thousands or millions of documents, it is impossible to search every document manually.

Therefore indexing becomes essential.

The indexing process is important because it

- * organizes information systematically
- * Makes searching faster
- * Improves retrieval accuracy
- * Reduces search time
- * Helps users find relevant documents easily.
- * Reduces duplicate searching
- * Improves the overall performance of an Information Retrieval System.

Decisions made During the Indexing process:-

(According to the textbook) Before creating an index, the indexer has to make some important decisions

1. selection of index terms
2. Decide the scope of indexing
3. Linking Index terms.

1) selection of Index terms:-

The first decision is to identify important concepts from the document. The selected words should represent the main topic of the document.

Ex:- Document title :- Applications of Artificial Intelligence in Healthcare

Possible Index terms

- * Artificial Intelligence
- * Health Care

* Machine Learning

* Diagnosis

* Medical Imaging

Words like the, is, are, of, for, are not selected because they do not describe the document.

Importance:

→ Improves searching

→ Reduces unnecessary index entries

→ Makes retrieval more accurate:

2) Beside the scope of Indexing:-

The indexer must decide how much of the document should be indexed.

The scope depends on:

* purpose of the index

* Type of document

* users requirements

* available time

* cost

for example some systems index only the title, while others index the entire document.

3) Linking Index Terms:- Many documents contain more than one concept.

Ex:- These concepts should be linked together

Document:- "Artificial Intelligence in Agriculture using IOT"

Index terms:- * Artificial Intelligence

* Agriculture

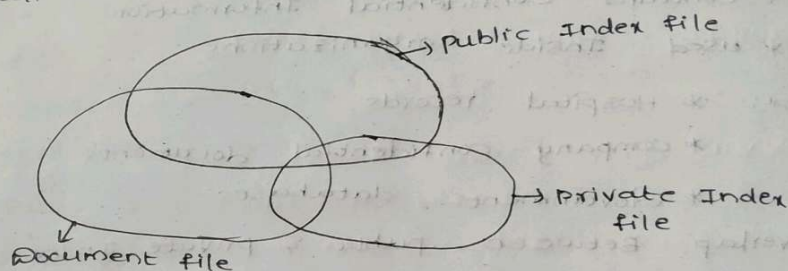
* IOT

- * Sensors
- * Smart farming

Advantages:-

- * connects related concepts
- * improve searching
- * improve Recall

If a user searches for IOT, the same document should still appear because all concepts are linked.



Document file:- The document file contains the complete original documents.

- Ex:-
- ⇒ Books
 - ⇒ Journals
 - ⇒ articles
 - ⇒ research papers
 - ⇒ PDF files

This is the primary source of information

2) public Index file:- The public Index file stores common Index terms that are available to all users.

features:-

- * accessible by everyone
- * used for public searching
- * contains general concepts

Ex:- * Google search Index
* Library catalogue

3) private Index file:-
The private Index file stores restricted or confidential Index terms.

Features:-

- * Accessible only to authorized users.
- * Contains Confidential Information.
- * used Inside organizations.

Ex:- * Hospital records
* company confidential documents
* Government databases.

overlap Between public & private Index files:-

The figure shows overlapping circles
This overlap indicates that

* some Keywords are available in both public & private Index files.

* some keywords are available only in public Index.

* some Keywords are available only in private Index.

This helps different users retrieve information according to their access rights.

Scope of Indexing:-

It refers to deciding how much of a document should be indexed and which concepts should be included in the index. Different indexers may create different indexes because:

- Their knowledge is different
- Their experience is different
- Their understanding of document is different.

Therefore, Indexing quality depends on the skill of the indexer.

Two Important factors in Scope of Indexing

There are two major factors

- 1) Exhaustivity of indexing
- 2) Specificity of indexing

1) Exhaustivity of indexing:-

• Exhaustivity means the no. of concepts selected from a documents.

- It answers the question
- How many topics from this document should be indexed
- High Exhaustivity means many topics from this document should be indexed.

Ex:- Document = "Microprocessor Architecture"

Index Terms:-

- Microprocessor
- Cache memory

- CPU
- Registers
- ALU
- control unit
- Instruction cycle
- memory hierarchy

Low Exhaustivity :-

only a few concepts are indexed

Example only • Microprocessor

• CPU

Advantages:-

- Small index
- faster indexing
- Less storage
- Lower cost

2. specificity of indexing:-

- Specificity means selecting how detailed or precise the index terms should be
- It answers the question
- "How specific should the keywords be?"

Ex: General term

Computer

Specific term

micro computer

highly specific term

pentium processor

The more specific the term the higher the precision.

High specificity advantages:-

- * High precision
- * Accurate search results
- * Less unwanted information

disadvantages:-
Some relevant documents may not be retrieved.

Low specificity advantages:-

- Broader search
- More documents retrieved

Disadvantages:-

- * Lower precision
- * More unwanted documents

Relationship b/w exhaustivity & specificity:-

High exhaustivity → High Recall

Low exhaustivity → Low Recall

High specificity → High precision

Low specificity → Low precision

weighting of Index Terms:-

weighting means assigning an importance value (weight) to every index term

The weight represents how important the concept is within the document

* Automatic Indexing:-

Introduction:- In an information retrieval system documents such as books, journals, articles reports, webpages. To retrieve these

documents quickly. They must be indexed.

Earlier Indexing was performed manually by human experts. However, due to the rapid growth of digital information, manual indexing became slow, expensive and difficult to maintain.

To overcome these problems, Automatic Indexing was introduced.

Automatic Indexing is a computerized process in which software programs automatically identify important words or concepts from documents and create an index without human intervention.

Today almost all modern search engines and digital libraries use automatic indexing.

Automatic Indexing:- It is the process of automatically selecting important keywords or index terms from documents using computer programs or algorithms and storing them in an index for efficient information retrieval.

Need for Automatic Indexing:- As the amount of digital information increases, everyday manual indexing becomes impractical.

Automatic Indexing is required because millions of new documents are created every day.

* Manual Indexing requires more time
* Human indexing is expensive
* Human errors are possible
* Human indexing provides faster
⇒ automatic indexing
Searching.

⇒ It supports online search engines

⇒ It can easily update indexes

Objectives of Automatic indexing:-

The main objectives are:

- * Reduce manual work
- * Increase search performance
- * Increase indexing speed
- * Handle very large document collections
- * Maintain consistency
- * Reduce indexing cost
- * Support quick retrieval of information
- * Improve user satisfaction.

Characteristics of Automatic Indexing:-

A good automatic indexing system should

- * work automatically
- * select meaningful keywords
- * ignore unnecessary words
- * produce consistent results
- * Support large databases
- * update indexes automatically
- * Retrieve documents quickly
- * Reduce storage space

Components of Automatic Indexing:-

1. Document Collection
2. Text Extraction
3. Stopword Removal
4. Normalization
5. Keyword selection
6. Assign index terms
7. Store in index file

1) Document Collection:- The system collects documents from various sources

Ex:- Books

articles

Journals

Webpages

Research papers

PDF files

2) Text Extraction:- The computer extracts all text from the document.

Ex:- Artificial Intelligence improves Healthcare using machine Learning

3) Stopword Removal:-

Common words that do not contribute to searching are removed.

ex:- the, is, are, in, on, with, of,

Remaining words become more meaningful

4) Normalization:- different forms of the same word are converted into one standard form.

5) Keyword selection:-
The computer identifies the most important words.

Eg sentence:- Artificial intelligence improves Healthcare using machine Learning

keywords:- * Artificial intelligence
* Healthcare
* Machine Learning

6) Assign Index Terms:-
The selected keywords are connected with the document.

ex:-

keyword	Document
AI	Doc 1
Healthcare	Doc 1
machine Learning	Doc 1

7) store in index file:- finally all keywords are stored in an index file whenever a user searches.

The search engine searches only the Index file instead of every document

Result:- fast Retrieval

Automatic Indexing process:-

plain text

Document collection



Text Extraction



Remove stopwords



Normalization



Keyword selection

Methods of Automatic Indexing

- 1) Indexing by term
- 2) Indexing by concept
- 3) Multimedia Indexing

1) Indexing by term:-

Indexing by term is a method of the automatic indexing in which the original words (terms) present in the document are used as index terms. The system extracts important words from the document and stores them in the index for retrieval.

When terms of the original document are used as the basis of indexing, two major techniques are used.

1) Statistical Techniques

2) Natural Language Techniques

1) Statistical Techniques:-

It creates indexes using mathematical & statistical information such as

- * frequency of occurrence of words
- * Distribution of words
- * weight of index terms

These techniques mainly include:

- * vector space model
- * probabilistic model
- * Bayesian model

vector space model (VSM):-

The VSM is one of the most widely used statistical models in information Retrieval.

It became popular through the smart Information Retrieval system developed by Dr. Gerard Salton at Cornell University.

In this model

→ Each document is represented as a vector.

→ Each unique word (term/token) forms one dimension of the vector.

→ Every term is assigned a weight.

→ If a word is absent its weight is 0.

Example:- vocabulary

AI
machine
Learning
Computer
Database

Document :- D₁

AI machine

Learning

vector Representation

AI = 1

machine = 1

Learning = 1

Computer = 0

Database = 0

Document vector (1, 1, 1, 0, 0)

Query is "AI Learning"

Query vector $1, 0, 1, 0, 0$

The System compares the query vector with Document vector.

If similarity is high, the document receives a higher rank.

2) probabilistic model:- Another statistical technique is the probabilistic model.

Instead of vectors, it calculates the probability that a document is relevant to the user's query.

Documents with higher probability receive higher ranks

3) Bayesian model:- The Bayesian model is a special probabilistic model based on Bayes's theorem.

It uses a Bayesian N/w to represent the relationship b/w concepts & features (words).

The model calculates:-

- * prior probability $p(c_i)$

- * Conditional probability $P_c(F_{ij}/c_i)$

- * posterior probability $p(c_i/F_{ij})$

Suppose you have 100 documents in your database out of these:

- * 40 documents are about Artificial Intelligence (AI).

- * 60 documents are about other topics

⇒ Now a document contains the word "Neural Network".

The system wants to answer this question "Is this document really about AI?"

To answer this, the Bayesian model calculates three probabilities.

1) prior probability: $P(C_i)$:- It means probability before looking at any words (features) in the document.

It is the initial belief that a document belongs to a concept.

Here

concept $(C_i) = \text{AI}$

out of 100 documents

40 are AI documents

So,

$$P(C_i) = 40/100 = 0.4 (40\%)$$

Prior probability is the probability that a document belongs to a concept before observing any features or words

2) conditional probability $P(F_{ij}/C_i)$:-

It means probability of seeing a feature (word) if the document already belongs to the concept.

Suppose among the 40 AI documents 30 contains the word "Neural Network".

$$\begin{aligned} \text{Then } P(\text{Neural Network} / \text{AI}) &= 30/40 = 0.75 \\ &= 75\% \end{aligned}$$

This means if a document is about AI, there is a 75% chance that it contains the word "Neural Network".

* Conditional probability is the probability that a feature (word) appears in a document given that the document belongs to a particular concept.

3) posterior probability - $P(C_i/F_{ij})$:-

Now the system sees a document containing the word "Neural Nlw"

It asks

"Since this word is present, what is the probability that the document is about AI?"

- This is called posterior probability
- It is calculated using Baye's theorem

Suppose the result is

$$P(AI/Neural\ Nlw) = 0.86 (86\%)$$

This means:

Since the word "Neural nlw" is present there is an 86% chance that the document is about AI.

The search engine therefore gives this document a higher rank.

⇒ posterior probability is the probability that a document belongs to a concept after observing the features (words) in the document.

Step 1:- Prior probability

↓
What do we know before reading the document.

Step 2:- Conditional probability

↓
How often does this word appear in AI documents.

Step 3:- posterior probability

↓
After seeing the word, how sure are we that this document is about AI?

* The posterior probability is used to rank documents.

B. Natural Language Techniques:-

Besides statistical methods indexing by term also uses Natural Language processing (NLP).

NLP performs deeper Language Analysis

Such as:

- * Morphological Analysis

- * Lexical Analysis

- * Syntactic Analysis

- * Semantic Analysis

- * Discourse Analysis

* Morphological Analysis:- Studies the structure and root form of words

* Lexical Analysis:- Identifies words, their

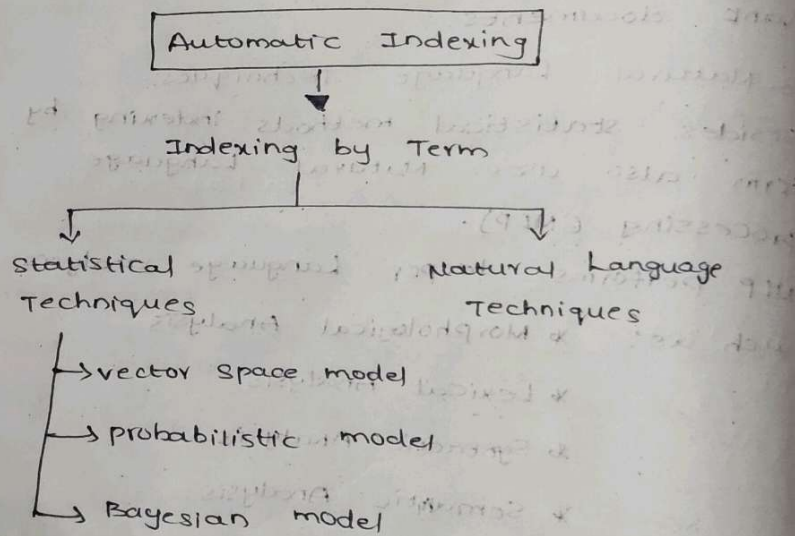
meanings about parts of speech (noun, verb, adjective, etc).

* Syntactic Analysis:- Analyzes the grammatical structure of a sentence.

* Semantic Analysis:- Determine the actual meaning of words & sentences based on context.

* Discourse Analysis:- Analyzes the relationship and meaning across multiple sentences in a document.

⇒ unlike statistical methods, NLP identifies the actual meaning (concepts) present in the document.



Indexing by Concept:-

Instead of indexing exact words the slm indexes concepts.

Different words with the same meaning are grouped together.

Ex:- Car
Automobile
Vehicle

All represent the same concept.

Multimedia Indexing:- It is the process of automatically or manually indexing multimedia data such as images, audio, ~~video~~ video, and graphics using their features metadata, or content to enable efficient Searching & retrieval.